

Experiment No. 3

Creating a sample database (Library Database System)

using any RDBMS package e.g. MySQL.

Use of DDL commands/queries

```
create database Library_DB_2025_26;
```

```
use Library_DB_2025_26;
```

```
select database();
```

Create a table - Publisher

```
create table Publisher (name varchar (40) ,
```

```
                        address varchar (30) ,
```

```
                        phone int);
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > show tables;
```

```
+-----+
| | Tables_in_library_db_2025_26 |
+-----+
| publisher |
+-----+
```

```
1 row in set (0.0014 sec)
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > desc
publisher;
```

```
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
```

```
+-----+-----+-----+-----+-----+-----+
| name | varchar(40) | YES | | NULL | |
```

```
+-----+-----+-----+-----+-----+-----+
| address | varchar(30) | YES | | NULL | |
```

```
+-----+-----+-----+-----+-----+-----+
| phone | int | YES | | NULL | |
```

```
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.0015 sec)
```

Add primary key constraint

```
alter table Publisher
```

```
add primary key (name);
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > desc
publisher;
```



```
#Add Referential Integrity constraint (i.e. Foreign key
constraint)
alter table Book
add foreign key (publisher_name) references publisher (name) on
delete cascade;
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > desc book;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(40)	YES		NULL	
pub_year	varchar(20)	YES		NULL	
publisher_name	varchar(40)	YES	MUL	NULL	

4 rows in set (0.0014 sec)

```
# Create a table - Book_Authors
create table Book_Authors (book_id int,
                           author_name varchar (30),
                           primary key (book_id, author_name),
                           foreign key (book_id) references book (book_id) on
delete cascade);
```

```
# Create a table - Library_Branch
create table Library_Branch (branch_id int primary key,
                             branch_name varchar (40),
                             address varchar (30));
```

```
#Add unique constraint
```

```
alter table Library_Branch
change branch_name branch_name varchar (40) not null unique;
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > desc
library_branch;
```

Field	Type	Null	Key	Default	Extra
branch_id	int	NO	PRI	NULL	
branch_name	varchar(40)	NO	UNI	NULL	
address	varchar(30)	YES		NULL	

3 rows in set (0.0015 sec)

```
# Create a table - Book_Copies
create table Book_Copies (book_id int,
                          branch_id int,
                          no_of_copies int,
                          primary key (book_id, branch_id),
foreign key (book_id) references book (book_id) on delete cascade,
foreign key (branch_id) references library_branch (branch_id) on
delete cascade);
```

```
#Add check constraint
alter table Book_Copies
add check (no_of_copies>0);
```

```
# Create the table - Borrower
create table Borrower
    (card_no int primary key,
     name varchar (30) not null,
     address varchar (50),
     phone bigint);
```

```
# Create the table - Book_Loans
create table Book_Loans
    (book_id int,
     branch_id int,
     card_no int not null,
     date_out date,
     due_date date,
     primary key (book_id, branch_id, card_no),
foreign key (book_id) references book (book_id) on delete cascade,
foreign key (branch_id) references library_branch (branch_id) on
delete cascade,
foreign key(card_no) references Borrower (card_no) on delete
cascade);
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > show tables;
```

```
+-----+
| Tables_in_library_db_2025_26 |
+-----+
| book |
| book_authors |
| book_copies |
| book_loans |
| borrower |
| library_branch |
| publisher |
+-----+
7 rows in set (0.0011 sec)
```

```
# To rename a table
```

```
rename table Borrower to Borrower1;
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > show tables;
```

```
+-----+
| Tables_in_library_db_2025_26 |
+-----+
| book                          |
| book_authors                  |
| book_copies                   |
| book_loans                    |
| borrower1                     |
| library_branch                |
| publisher                     |
+-----+
```

```
7 rows in set (0.0013 sec)
```

```
rename table Borrower1 to Borrower;
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > show tables;
```

```
+-----+
| Tables_in_library_db_2025_26 |
+-----+
| book                          |
| book_authors                  |
| book_copies                   |
| book_loans                    |
| borrower                      |
| library_branch                |
| publisher                     |
+-----+
```

```
7 rows in set (0.0049 sec)
```

```
#drop database Library_DB_2025_26;
```

```
#drop table publisher;
```

Experiment No. 4

SQL DML queries: Use of SQL DML queries to retrieve the data,

insert the data, delete the record and update the database.

```
use Library_DB_2025_26;
select database();
```

To insert records into Publisher table

```
insert into Publisher values ('Mcgraw-Hill', 'Bangalore', 9989076587);
insert into Publisher values ('Pearson', 'NewDelhi', 9889076565);
insert into Publisher values ('Random House', 'Hyderabad', 7455679345);
insert into Publisher values ('Hachette Livre', 'Chennai', 8970862340);
insert into Publisher values ('Grupo Planeta', 'Bangalore', 7756120238);
insert into Publisher values ('Wiley India', 'New Delhi', 1143630000);
insert into Publisher values ('S Chand', 'Noida', 7291975264);
insert into Publisher values ('Technical Publications', 'Pune', 9763719464);
insert into Publisher values ('Nirali Prakashan', 'Pune', 2025512336);
insert into Publisher values ('Prakash Publication', 'Jalgaon', 9272514482);
insert into Publisher values ('Galgotia Publications', 'New Delhi', 1123263334);
```

MySQL localhost:3306 ssl library_db_2025_26 SQL > select * from publisher;

name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Nirali Prakashan	Pune	2025512336
Pearson	New Delhi	9889076565
Prakash Publication	Jalgaon	9272514482
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Technical Publications	Pune	9763719464
Wiley India	New Delhi	1143630000

11 rows in set (0.0115 sec)

To insert records into Book table

```
insert into book values (1,'DBMS','2017', 'Mcgraw-Hill');
insert into book values (2,'ADBMS','2016', 'Mcgraw-Hill');
insert into book values (3,'CN','2016', 'Pearson');
insert into book values (4,'CG','2015', 'Grupo Planeta');
insert into book values (5,'OS','2016', 'Pearson');
insert into book values (6,'CO','2008', 'Nirali Prakashan');
insert into book values (7,'CO','2005', 'Galgotia Publications');
insert into book values (8,'DBMS','2022', 'Technical Publications');
insert into book values (9,'DBMS for Distributed System','2022', 'Pearson');
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > select * from Book;
```

book_id	title	pub_year	publisher_name
1	DBMS	2017	Mcgraw-Hill
2	ADBMS	2016	Mcgraw-Hill
3	CN	2016	Pearson
4	CG	2015	Grupo Planeta
5	OS	2016	Pearson
6	CO	2008	Nirali Prakashan
7	CO	2005	Galgotia Publications
8	DBMS	2022	Technical Publications
9	DBMS for Distributed System	2022	Pearson

```
9 rows in set (0.0006 sec)
```

```
# To insert records into Book_Authors table
```

```
insert into book_authors values (1, 'Navathe');
```

```
insert into book_authors values (2, 'Navathe');
```

```
insert into book_authors values (3, 'Tanenbaum');
```

```
insert into book_authors values (4, 'Edward Angel');
```

```
insert into book_authors values (5, 'Galvin');
```

```
insert into book_authors values (6, 'K. P. Adhiya');
```

```
insert into book_authors values (7, 'Navin Kumar');
```

```
insert into book_authors values (8, 'A. A. Puntambekar');
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL > select * from Book_Authors;
```

book_id	author_name
1	Navathe
2	Navathe
3	Tanenbaum
4	Edward Angel
5	Galvin
6	K. P. Adhiya
7	Navin Kumar
8	A. A. Puntambekar

```
8 rows in set (0.0008 sec)
```

```
# To insert records into Library_Branch table
```

```
insert into library_branch values (10,'RR Nagar','Bangalore');
```

```
insert into library_branch values (11,'RNSIT','Bangalore');
```

```
insert into library_branch values (12,'Rajaji Nagar','Bangalore');
```

```
insert into library_branch values (13,'NITTE','Mangalore');
```

```
insert into library_branch values (14,'Manipal','Udupi');
```

```
insert into library_branch values (15,'SSBT','Jalgaon');
```

```
insert into library_branch values (16,'SPIT','Mumbai');
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL> select * from Library_Branch;
```

branch_id	branch_name	address
10	RR Nagar	Bangalore
11	RNSIT	Bangalore
12	Rajaji Nagar	Bangalore
13	NITTE	Mangalore
14	Manipal	Udupi
15	SSBT	Jalgaon
16	SPIT	Mumbai

```
7 rows in set (0.0008 sec)
```

```
# To insert records into Book_Copies table
```

```
insert into book_copies values (1, 10, 10);
insert into book_copies values (1, 11, 5);
insert into book_copies values (2, 12, 2);
insert into book_copies values (2, 13, 5);
insert into book_copies values (3, 14, 7);
insert into book_copies values (5, 10, 1);
insert into book_copies values (4, 11, 3);
insert into book_copies values (6, 15, 13);
insert into book_copies values (6, 16, 10);
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL> select * from Book_Copies;
```

book_id	branch_id	no_of_copies
1	10	10
1	11	5
2	12	2
2	13	5
3	14	7
4	11	3
5	10	1
6	15	13
6	16	10

```
9 rows in set (0.0008 sec)
```

```
# To insert records into Borrower table
```

```
insert into borrower values (100, 'George', 'Jalgaon', 1122334455 );
insert into borrower values (101, 'Krishna', 'Daryapur', 9421516499 );
insert into borrower values (102, 'Uday', 'Dhule', 3322445566 );
insert into borrower values (103, 'Nilesh', 'Nagpur', 7722445566 );
insert into borrower values (104, 'Abhijit', 'Amravati', 7777445566 );
insert into borrower values (105, 'Hitesh', 'Amravati', 8888445566 );
```



```
MySQL localhost:3306 ssl library_db_2025_26 SQL> select * from
borrower;
```

card_no	name	address	phone
100	George	Jalgaon	1122334455
101	Krishna	Daryapur	9421516499
102	Uday	Dhule	3322445566
103	Nilesh	Nagpur	7722445566
104	Abhijit	Amravati	7777445566
105	Hitesh	Amravati	8888445566

```
6 rows in set (0.0006 sec)
```

```
# To insert records into Book_Loans table
```

```
insert into book_loans values (1, 10, 101, '2017-01-10', '2017-06-17');
insert into book_loans values (3, 14, 101, '2017-03-18', '2017-07-17');
insert into book_loans values (2, 13, 101, '2017-02-11', '2017-04-21');
insert into book_loans values (4, 11, 101, '2017-03-15', '2017-07-15');
insert into book_loans values (1, 11, 104, '2017-04-12', '2017-05-12');
```

```
MySQL localhost:3306 ssl library_db_2025_26 SQL> select * from
Book_Loans;
```

book_id	branch_id	card_no	date_out	due_date
1	10	101	2017-01-10	2017-06-17
1	11	104	2017-04-12	2017-05-12
2	13	101	2017-02-11	2017-04-21
3	14	101	2017-03-18	2017-07-17
4	11	101	2017-03-15	2017-07-15

```
5 rows in set (0.0006 sec)
```

```
# List the names of all publishers
```

```
select name
from Publisher;
```

name
Galgotia Publications
Grupo Planeta
Hachette Livre
Mcgraw-Hill
Nirali Prakashan
Pearson

```

| Prakash Publication |
| Random House       |
| S Chand             |
| Technical Publications |
| Wiley India         |
+-----+
11 rows in set (0.0023 sec)

```

```

# List the names of all publishers with contact number
select name, phone
from Publisher;

```

```

+-----+-----+
| name                | phone      |
+-----+-----+
| Galgotia Publications | 1123263334 |
| Grupo Planeta         | 7756120238 |
| Hachette Livre        | 8970862340 |
| Mcgraw-Hill           | 9989076587 |
| Nirali Prakashan      | 2025512336 |
| Pearson               | 9889076565 |
| Prakash Publication   | 9272514482 |
| Random House          | 7455679345 |
| S Chand               | 7291975264 |
| Technical Publications | 9763719464 |
| Wiley India           | 1143630000 |
+-----+-----+
11 rows in set (0.0007 sec)

```

```

# List names of all publishers from New Delhi
select name
from Publisher
where address = 'New Delhi';

```

```

+-----+
| name                |
+-----+
| Galgotia Publications |
| Wiley India          |
+-----+
3 rows in set (0.0007 sec)

```

```

#Use of update
update Publisher
set address = 'New Delhi'
where name = 'Pearson';

```

```
select name
from Publisher
where address = 'New Delhi';
```

```
+-----+
| name          |
+-----+
| Galgotia Publications |
| Pearson        |
| Wiley India    |
+-----+
3 rows in set (0.0007 sec)
```

```
# List the titles of books published by publishers
select title
from Book;
```

```
+-----+
| title          |
+-----+
| DBMS           |
| ADBMS          |
| CN             |
| CG             |
| OS             |
| CO             |
| CO             |
| DBMS           |
| DBMS for Distributed System |
+-----+
9 rows in set (0.0022 sec)
```

```
select distinct title
from Book;
```

```
+-----+
| title          |
+-----+
| DBMS           |
| ADBMS          |
| CN             |
| CG             |
| OS             |
| CO             |
| DBMS for Distributed System |
+-----+
7 rows in set (0.0006 sec)
```

Display the details of books, alphabetically titled

```
select *  
from Book  
order by title;
```

book_id	title	pub_year	publisher_name
2	ADBMS	2016	Mcgraw-Hill
4	CG	2015	Grupo Planeta
3	CN	2016	Pearson
6	CO	2008	Nirali Prakashan
7	CO	2005	Galgotia Publications
1	DBMS	2017	Mcgraw-Hill
8	DBMS	2022	Technical Publications
9	DBMS for Distributed System	2022	Pearson
5	OS	2016	Pearson

9 rows in set (0.0007 sec)

```
select title  
from Book  
order by title desc;
```

title
OS
DBMS for Distributed System
DBMS
DBMS
CO
CO
CN
CG
ADBMS

9 rows in set (0.0007 sec)

#Find the latest publication year for DBMS book

```
select title, pub_year  
from Book  
where title = 'DBMS'  
order by pub_year desc  
limit 1;
```

title	pub_year
DBMS	2022

1 row in set (0.0104 sec)

```
# Display the details of books, publication year wise
select *
from Book
order by pub_year;
```

```
select title, pub_year
from Book
order by pub_year desc;
```

```
+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                  | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                  | 2017     |
| ADBMS                 | 2016     |
| CN                    | 2016     |
| OS                    | 2016     |
| CG                    | 2015     |
| CO                    | 2008     |
| CO                    | 2005     |
+-----+-----+
9 rows in set (0.0007 sec)
```

```
# To delete records
select * from book_copies;
delete from book_copies
where (book_id=1 and branch_id=10)
or (book_id=1 and branch_id=11)
or (book_id=2 and branch_id=12);
select * from book_copies;
```

```
insert into book_copies values (1, 10, 10);
insert into book_copies values (1, 11, 5);
insert into book_copies values (2, 12, 2);
select * from book_copies;
```

```
# List title of books and publication year, published between 2015
to 2022. Display recent publishing year on top.
```

```
select title, pub_year
from Book
where pub_year between '2015' and '2022'
order by pub_year desc;
```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                 | 2017     |
| ADBMS                | 2016     |
| CN                   | 2016     |
| OS                   | 2016     |
| CG                   | 2015     |
+-----+-----+

```

7 rows in set (0.0005 sec)

List title of books and publication year, published between 2017 to 2022.

```

select title, pub_year
from Book
where pub_year between '2017' and '2022';

```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2017     |
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
+-----+-----+

```

3 rows in set (0.0006 sec)

Find out average number of copies of each book

```

select book_id, avg(no_of_copies) as Average_No_of_Copies_Per_Book
from Book_Copies
group by book_id;

```

```

+-----+-----+
| book_id | Average_No_of_Copies_Per_Book |
+-----+-----+
| 1       | 7.5000                        |
| 2       | 3.5000                        |
| 3       | 7.0000                        |
| 4       | 3.0000                        |
| 5       | 1.0000                        |
| 6       | 11.5000                       |
+-----+-----+

```

6 rows in set (0.0010 sec)

Display the details of books along with average number of copies of each book

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      1 | Navathe    | DBMS  |              7.5000 |
|      2 | Navathe    | ADBMS |              3.5000 |
|      3 | Tanenbaum  | CN    |              7.0000 |
|      4 | Edward Angel | CG    |              3.0000 |
|      5 | Galvin     | OS    |              1.0000 |
|      6 | K. P. Adhiya | CO    |             11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0010 sec)
```

Find out Authors whose average no of book copies in library are more than 3 and display the name of author on top having maximum average number of book copies.

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id,author_name, title
having Average_No_of_Copies_Per_Book >3
order by Average_No_of_Copies_Per_Book desc;
```

```
+-----+-----+-----+-----+
-+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
-+
|      6 | K. P. Adhiya | CO    |             11.5000 |
|      1 | Navathe    | DBMS  |              7.5000 |
|      3 | Tanenbaum  | CN    |              7.0000 |
|      2 | Navathe    | ADBMS |              3.5000 |
+-----+-----+-----+-----+
-+
4 rows in set (0.0008 sec)
```

```
# Display - title, publication year and publisher name for all titles
which includes DBMS.
```

```
select title, pub_year, publisher_name
from Book
where title like '%DBMS%';
```

```
+-----+-----+-----+
| title                | pub_year | publisher_name      |
+-----+-----+-----+
| DBMS                 | 2017     | McGraw-Hill         |
| ADBMS                | 2016     | McGraw-Hill         |
| DBMS                 | 2022     | Technical Publications |
| DBMS for Distributed System | 2022     | Pearson              |
+-----+-----+-----+
4 rows in set (0.0006 sec)
```

```
# Display the details of publishers from Pune and Jalgaon.
```

```
select *
from publisher
where address = 'Pune' or address = 'Jalgaon';
```

```
select *
from publisher
where address in ('Pune', 'Jalgaon');
```

```
+-----+-----+-----+
| name                | address  | phone              |
+-----+-----+-----+
| Nirali Prakashan    | Pune     | 2025512336        |
| Prakash Publication | Jalgaon  | 9272514482        |
| Technical Publications | Pune     | 9763719464        |
+-----+-----+-----+
3 rows in set (0.0045 sec)
```

```
#Display the details of publishers other than Pune and Jalgaon.
```

```
select *
from publisher
where address not in ('Pune', 'Jalgaon');
```


name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Pearson	New Delhi	9889076565
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Wiley India	New Delhi	1143630000

8 rows in set (0.0006 sec)

Use of IS NULL, IS NOT NULL

```
select *
from publisher
where phone is null;
```

```
select *
from publisher
where phone is not null;
```

name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Nirali Prakashan	Pune	2025512336
Pearson	New Delhi	9889076565
Prakash Publication	Jalgaon	9272514482
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Technical Publications	Pune	9763719464
Wiley India	New Delhi	1143630000

11 rows in set (0.0007 sec)

Experiment No. 5

To use various MySQL Commands/Queries such as - use of Aggregate Functions e.g. SUM(), COUNT(), AVG() etc., GROUP BY clause, HAVING clause, ORDER BY clause etc.

```
use Library_DB_2025_26;
select database();
```

```
# Find out average number of copies of each book
select book_id, avg(no_of_copies) as Average_No_of_Copies_Per_Book
from Book_Copies
group by book_id;
```

```
+-----+-----+
| book_id | Average_No_of_Copies_Per_Book |
+-----+-----+
|      1 |              7.5000 |
|      2 |              3.5000 |
|      3 |              7.0000 |
|      4 |              3.0000 |
|      5 |              1.0000 |
|      6 |             11.5000 |
+-----+-----+
6 rows in set (0.0010 sec)
```

Display the details of books along with average number of copies of each book

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      1 | Navathe    | DBMS  |              7.5000 |
|      2 | Navathe    | ADBMS |              3.5000 |
|      3 | Tanenbaum  | CN    |              7.0000 |
|      4 | Edward Angel | CG    |              3.0000 |
|      5 | Galvin     | OS    |              1.0000 |
|      6 | K. P. Adhiya | CO    |             11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0010 sec)
```

Find out Authors whose average no of book copies in library are more than 3 and display the name of author on top having maximum average number of book copies.

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id,author_name, title
having Average_No_of_Copies_Per_Book >3
order by Average_No_of_Copies_Per_Book desc;
```

```
+-----+-----+-----+-----+
-+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
-+
|      6 | K. P. Adhiya | CO      | 11.5000 |
|      1 | Navathe      | DBMS    | 7.5000  |
|      3 | Tanenbaum    | CN      | 7.0000  |
|      2 | Navathe      | ADBMS   | 3.5000  |
+-----+-----+-----+-----+
-+
4 rows in set (0.0008 sec)
```

```
# Find out Author's name who wrote maximum books
select author_name, count(author_name) as Number_of_Books_Written
from Book_Authors
group by author_name
order by Number_of_Books_Written desc
limit 1;
```

```
+-----+-----+
| author_name | Number_of_Books_Written |
+-----+-----+
| Navathe      | 2 |
+-----+-----+
1 row in set (0.0009 sec)
```

```
# Find out total number of Publishers
select count(*) as Total_No_of_Publishers
from Publisher;
```

```
+-----+
| Total_No_of_Publishers |
+-----+
| 11 |
+-----+
1 row in set (0.0007 sec)
```

Proper Query: - Find out the book and its corresponding branch having maximum number of copies

```
select *
from book_copies
where no_of_copies =
      (select max(no_of_copies)
       from book_copies);
```

```
+-----+-----+-----+
| book_id | branch_id | no_of_copies |
+-----+-----+-----+
|      6 |      15 |      13 |
+-----+-----+-----+
1 row in set (0.0009 sec)
```

```
select *
from book_copies
where no_of_copies =
      (select min(no_of_copies)
       from book_copies);
```

```
+-----+-----+-----+
| book_id | branch_id | no_of_copies |
+-----+-----+-----+
|      5 |      10 |      1 |
+-----+-----+-----+
1 row in set (0.0007 sec)
```

Display total number of copies of all books

```
select sum(no_of_copies) as Total_Copies_of_All_Books
from Book_Copies;
```

```
+-----+
| Total_Copies_of_All_Books |
+-----+
|              56 |
+-----+
1 row in set (0.0006 sec)
```

List title of books and publication year, published between 2015 to 2022. Display recent publishing year on top.

```
select title, pub_year
from Book
where pub_year between '2015' and '2022'
order by pub_year desc;
```

```

+-----+-----+
| title                                | pub_year |
+-----+-----+
| DBMS                                | 2022     |
| DBMS for Distributed System         | 2022     |
| DBMS                                | 2017     |
| ADBMS                              | 2016     |
| CN                                  | 2016     |
| OS                                  | 2016     |
| CG                                  | 2015     |
+-----+-----+

```

7 rows in set (0.0005 sec)

List title of books and publication year, published between 2017 to 2022.

```

select title, pub_year
from Book
where pub_year between '2017' and '2022';

```

```

+-----+-----+
| title                                | pub_year |
+-----+-----+
| DBMS                                | 2017     |
| DBMS                                | 2022     |
| DBMS for Distributed System         | 2022     |
+-----+-----+

```

3 rows in set (0.0006 sec)

Find out average number of copies of each book

```

select book_id, avg(no_of_copies) as Average_No_of_Copies_Per_Book
from Book_Copies
group by book_id;

```

```

+-----+-----+
| book_id | Average_No_of_Copies_Per_Book |
+-----+-----+
| 1       | 7.5000                         |
| 2       | 3.5000                         |
| 3       | 7.0000                         |
| 4       | 3.0000                         |
| 5       | 1.0000                         |
| 6       | 11.5000                       |
+-----+-----+

```

6 rows in set (0.0010 sec)

Find out Authors whose average no of book copies in library are more than 3 and display the name of author on top having maximum average number of book copies.

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id,author_name, title
having Average_No_of_Copies_Per_Book >3
order by Average_No_of_Copies_Per_Book desc;
```

book_id	author_name	title	Average_No_of_Copies_Per_Book
6	K. P. Adhiya	CO	11.5000
1	Navathe	DBMS	7.5000
3	Tanenbaum	CN	7.0000
2	Navathe	ADBMS	3.5000

4 rows in set (0.0008 sec)

Display the details of publishers from Pune and Jalgaon.

```
select *
from publisher
where address = 'Pune' or address = 'Jalgaon';
```

```
select *
from publisher
where address in ('Pune','Jalgaon');
```

name	address	phone
Nirali Prakashan	Pune	2025512336
Prakash Publication	Jalgaon	9272514482
Technical Publications	Pune	9763719464

3 rows in set (0.0045 sec)

#Display the details of publishers other than Pune and Jalgaon.

```
select *
from publisher
where address not in ('Pune','Jalgaon');
```

name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Pearson	New Delhi	9889076565
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Wiley India	New Delhi	1143630000

8 rows in set (0.0006 sec)

Experiment No. 6

#To use various MySQL Commands/Queries such as - Set
#Operations e.g. UNION, UNION ALL, Set Membership
#Operations e.g. IN, NOT IN, BETWEEN, Set Comparison
#Operations e.g. SOME (or ANY), ALL.

```
use Library_DB_2025_26;  
select database();
```

Display Publishers name from New Delhi and Pune cities, using UNION
operator

```
select name, address  
from Publisher  
where address = 'New Delhi'  
union  
select name, address  
from Publisher  
where address = 'Pune';
```

```
select name, address  
from Publisher  
where address = 'New Delhi'  
union all  
select name, address  
from Publisher  
where address = 'Pune';
```

```
+-----+-----+  
| name                | address  |  
+-----+-----+  
| Galgotia Publications | New Delhi |  
| Pearson              | New Delhi |  
| Wiley India          | New Delhi |  
| Nirali Prakashan     | Pune     |  
| Technical Publications | Pune     |  
+-----+-----+  
5 rows in set (0.0009 sec)
```



```
# Display the identification number of books and their corresponding
author name, using Book table and Book_Authors table
```

```
# Use of join on
```

```
select Book.book_id, Book_authors.book_id, author_name
from Book join Book_Authors
on Book.book_id = Book_Authors.book_id;
```

```
# Use of inner join on
```

```
select Book.book_id, Book_authors.book_id, author_name
from Book inner join Book_Authors
on Book.book_id = Book_Authors.book_id;
```

```
#Use of natural join
```

```
select Book.book_id, Book_authors.book_id, author_name
from Book natural join Book_Authors;
```

```
+-----+-----+-----+
| book_id | book_id | author_name |
+-----+-----+-----+
|      7 |      7 | Navin Kumar |
|      4 |      4 | Edward Angel |
|      1 |      1 | Navathe     |
|      2 |      2 | Navathe     |
|      6 |      6 | K. P. Adhiya |
|      3 |      3 | Tanenbaum   |
|      5 |      5 | Galvin      |
|      8 |      8 | A. A. Puntambekar |
+-----+-----+-----+
8 rows in set (0.0005 sec)
```

```
# Display the details of books along with average number of copies
of each book
```

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      1 | Navathe     | DBMS  | 7.5000 |
|      2 | Navathe     | ADBMS | 3.5000 |
|      3 | Tanenbaum   | CN    | 7.0000 |
|      4 | Edward Angel | CG    | 3.0000 |
|      5 | Galvin      | OS    | 1.0000 |
|      6 | K. P. Adhiya | CO    | 11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0006 sec)
```

```
# Display the details of publishers from Pune and Jalgaon.
```

```
select *
from publisher
where address = 'Pune' or address = 'Jalgaon';
```

```
select *
from publisher
where address in ('Pune','Jalgaon');
```

```
+-----+-----+-----+
| name                | address | phone      |
+-----+-----+-----+
| Nirali Prakashan    | Pune    | 2025512336 |
| Prakash Publication | Jalgaon | 9272514482 |
| Technical Publications | Pune    | 9763719464 |
+-----+-----+-----+
3 rows in set (0.0045 sec)
```

```
#Display the details of publishers other than Pune and Jalgaon.
```

```
select *
from publisher
where address not in ('Pune','Jalgaon');
```

```
+-----+-----+-----+
| name                | address | phone      |
+-----+-----+-----+
| Galgotia Publications | New Delhi | 1123263334 |
| Grupo Planeta        | Bangalore | 7756120238 |
| Hachette Livre       | Chennai  | 8970862340 |
| Mcgraw-Hill          | Bangalore | 9989076587 |
| Pearson              | New Delhi | 9889076565 |
| Random House         | Hyderabad | 7455679345 |
| S Chand              | Noida    | 7291975264 |
| Wiley India          | New Delhi | 1143630000 |
+-----+-----+-----+
8 rows in set (0.0006 sec)
```

```
# List title of books and publication year, published between 2015
to 2022. Display recent publishing year on top.
```

```
select title, pub_year
from Book
where pub_year between '2015' and '2022'
order by pub_year desc;
```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                 | 2017     |
| ADBMS                | 2016     |
| CN                   | 2016     |
| OS                   | 2016     |
| CG                   | 2015     |
+-----+-----+
7 rows in set (0.0005 sec)

```

List title of books and publication year, published between 2017 to 2022.

```

select title, pub_year
from Book
where pub_year between '2017' and '2022';

```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2017     |
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
+-----+-----+
3 rows in set (0.0006 sec)

```

```

select title, pub_year
from Book
where pub_year not between '2015' and '2022'
order by pub_year desc;

```

```

+-----+-----+
| title | pub_year |
+-----+-----+
| CO    | 2008     |
| CO    | 2005     |
+-----+-----+
2 rows in set (0.0004 sec)

```

Use of Set comparison operator - SOME/ANY operator

```
select book_id, branch_id
from book_copies
where no_of_copies >any
      (select no_of_copies
       from book_copies
       where book_id = 1);
```

```
+-----+-----+
| book_id | branch_id |
+-----+-----+
|      1 |      10 |
|      3 |      14 |
|      6 |      15 |
|      6 |      16 |
+-----+-----+
4 rows in set (0.0007 sec)
```

Use of Set comparison operator - ALL operator

```
select book_id, branch_id
from book_copies
where no_of_copies >all
      (select no_of_copies
       from book_copies
       where book_id = 1);
```

```
+-----+-----+
| book_id | branch_id |
+-----+-----+
|      6 |      15 |
+-----+-----+
1 row in set (0.0006 sec)
```

Experiment No. 7: Group B - 01

To use various MySQL Commands/Queries related to Join
operations, such as - INNER JOIN, LEFT JOIN, RIGHT JOIN,
Full Join using UNION.

```
use Library_DB_2025_26;  
select database();
```

Display the identification number of books and their corresponding
author name, using Book table and Book_Authors table

Use of join on

```
select Book.book_id, Book_authors.book_id, author_name  
from Book join Book_Authors
```

```
on Book.book_id = Book_Authors.book_id;
```

Use of inner join on

```
select Book.book_id, Book_authors.book_id, author_name  
from Book inner join Book_Authors
```

```
on Book.book_id = Book_Authors.book_id;
```

#Use of natural join

```
select Book.book_id, Book_authors.book_id, author_name  
from Book natural join Book_Authors;
```

```
+-----+-----+-----+  
| book_id | book_id | author_name |  
+-----+-----+-----+  
|      7 |      7 | Navin Kumar |  
|      4 |      4 | Edward Angel |  
|      1 |      1 | Navathe     |  
|      2 |      2 | Navathe     |  
|      6 |      6 | K. P. Adhiya |  
|      3 |      3 | Tanenbaum   |  
|      5 |      5 | Galvin      |  
|      8 |      8 | A. A. Puntambekar |  
+-----+-----+-----+
```

8 rows in set (0.0005 sec)

Display the details of books along with average number of copies
of each book

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)  
as Average_No_of_Copies_Per_Book  
from Book_Copies join Book_Authors join Book  
on Book_Copies.book_id = Book_Authors.book_id and Book.book_id =  
Book_Authors.book_id  
group by book_id, author_name, title;
```

book_id	author_name	title	Average_No_of_Copies_Per_Book
1	Navathe	DBMS	7.5000
2	Navathe	ADBMS	3.5000
3	Tanenbaum	CN	7.0000
4	Edward Angel	CG	3.0000
5	Galvin	OS	1.0000
6	K. P. Adhiya	CO	11.5000

6 rows in set (0.0006 sec)

```
# List title of those books who don't have any author
select title, author_name
from book left join book_authors
on book.book_id = book_authors.book_id
where book_authors.author_name is null;
```

```
select title, author_name
from book_authors right join book
on book.book_id = book_authors.book_id
where book_authors.author_name is null;
```

title	author_name
DBMS for Distributed System	NULL

1 row in set (0.0004 sec)

```
# Use of full join concept
select title
from book left join book_authors
on book.book_id = book_authors.book_id
where book_authors.author_name is null
union
select title
from book_authors right join book
on book.book_id = book_authors.book_id
where book_authors.author_name is null;
```

title
DBMS for Distributed System

1 row in set (0.0009 sec)

```

select title
from book left join book_authors
on book.book_id = book_authors.book_id
where book_authors.author_name is null
union all
select title
from book_authors right join book
on book.book_id = book_authors.book_id
where book_authors.author_name is null;

```

```

+-----+
| title |
+-----+
| DBMS for Distributed System |
| DBMS for Distributed System |
+-----+
2 rows in set (0.0006 sec)

```

```

# Display the details of books along with average number of copies
# of each book
# Use of equi-join

```

```

select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;

```

```

+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
| 1 | Navathe | DBMS | 7.5000 |
| 2 | Navathe | ADBMS | 3.5000 |
| 3 | Tanenbaum | CN | 7.0000 |
| 4 | Edward Angel | CG | 3.0000 |
| 5 | Galvin | OS | 1.0000 |
| 6 | K. P. Adhiya | CO | 11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0008 sec)

```

```

# Display the identification number of books and their corresponding
author name, using Book table and Book_Authors table
# Use of join on
select Book.book_id, Book_authors.book_id, author_name
from Book join Book_Authors
on Book.book_id = Book_Authors.book_id;

```

```
# Use of inner join on
select Book.book_id, Book_authors.book_id, author_name
from Book inner join Book_Authors
on Book.book_id = Book_Authors.book_id;
#Use of natural join
select Book.book_id, Book_authors.book_id, author_name
from Book natural join Book_Authors;
```

```
+-----+-----+-----+
| book_id | book_id | author_name |
+-----+-----+-----+
|      7 |      7 | Navin Kumar |
|      4 |      4 | Edward Angel |
|      1 |      1 | Navathe     |
|      2 |      2 | Navathe     |
|      6 |      6 | K. P. Adhiya |
|      3 |      3 | Tanenbaum   |
|      5 |      5 | Galvin      |
|      8 |      8 | A. A. Puntambekar |
+-----+-----+-----+
8 rows in set (0.0005 sec)
```


Experiment No. 8: Group B - 02

To create a trigger and write a program to demonstrate
its use in MySQL.

```
use Library_DB_2025_26;  
select database();
```

```
# Trigger  
# If number of copies of any book purchased for any branch is more  
# than 10, then as a discount, branch receives one more additional  
# copy. For more than 20, additional 2 copies and so on. (i.e. one  
# more additional copy per 10 books)
```

```
delimiter $$  
create trigger trigger_befor_insert_Lib_DB before insert  
on book_copies for each row  
begin  
    if new.no_of_copies>10  
    then  
        set new.no_of_copies = new.no_of_copies +  
                                floor(new.no_of_copies * 0.10);  
    end if;  
end $$  
delimiter ;
```

```
show triggers;
```

```
select * from book_copies;
```

```
+-----+-----+-----+  
| book_id | branch_id | no_of_copies |  
+-----+-----+-----+  
|      1 |      10 |           10 |  
|      1 |      11 |            5 |  
|      2 |      12 |            2 |  
|      2 |      13 |            5 |  
|      3 |      14 |            7 |  
|      4 |      11 |            3 |  
|      5 |      10 |            1 |  
|      6 |      15 |           13 |  
|      6 |      16 |           10 |  
+-----+-----+-----+  
9 rows in set (0.0009 sec)
```

```

insert into book_copies values(3, 15, 11);
select * from book_copies;
insert into book_copies values(3, 16, 17);
select * from book_copies;
insert into book_copies values(3, 13, 21);
select * from book_copies;

```

book_id	branch_id	no_of_copies
1	10	10
1	11	5
2	12	2
2	13	5
3	13	23
3	14	7
3	15	12
3	16	18
4	11	3
5	10	1
6	15	13
6	16	10

```

+-----+-----+-----+
12 rows in set (0.0005 sec)

```

```

delete from book_copies
where (book_id=3 and branch_id=15)
or (book_id=3 and branch_id=16)
or (book_id=3 and branch_id=13);

```

```

select * from book_copies;

```

```

drop trigger trigger_befor_insert_Lib_DB;

```

```

show triggers;

```

Experiment No. 9: Group B - 03

To create a View and to perform various operations on a
View in MySQL.

```
use Library_DB_2025_26;  
select database();
```

View

To create a view for students - To display his/her card number,
book id & book title against his/her name, branch id & branch name
from where book is issued and due date.

```
create view View_for_Student_Lib_DB as  
select card_no, book_loans.book_id, title, book_loans.branch_id,  
                                             branch_name, due_date  
from book_loans, book, library_branch  
where book_loans.branch_id = library_branch.branch_id and  
book_loans.book_id = book.book_id;
```

```
show full tables;
```

```
+-----+-----+  
| Tables_in_library_db_2025_26 | Table_type |  
+-----+-----+  
| book                          | BASE TABLE |  
| book_authors                  | BASE TABLE |  
| book_copies                   | BASE TABLE |  
| book_loans                    | BASE TABLE |  
| borrower                      | BASE TABLE |  
| library_branch                | BASE TABLE |  
| publisher                     | BASE TABLE |  
| view_for_student_lib_db       | VIEW        |  
+-----+-----+  
8 rows in set (0.0012 sec)
```

```
select * from View_for_Student_Lib_DB;
```

card_no	book_id	title	branch_id	branch_name	due_date
101	1	DBMS	10	RR Nagar	2017-06-17
104	1	DBMS	11	RNSIT	2017-05-12
101	2	ADBMS	13	NITTE	2017-04-21
101	3	CN	14	Manipal	2017-07-17
101	4	CG	11	RNSIT	2017-07-15

5 rows in set (0.0007 sec)

```
select *
from View_for_Student_Lib_DB
where card_no=104;
```

card_no	book_id	title	branch_id	branch_name	due_date
104	1	DBMS	11	RNSIT	2017-05-12

1 row in set (0.0010 sec)

```
# Update a View
update View_for_Student_Lib_DB
set due_date = '2017-07-12'
where card_no = 104;
```

```
select *
from View_for_Student_Lib_DB
where card_no=104;
select * from book_loans
where card_no=104;
```

card_no	book_id	title	branch_id	branch_name	due_date
104	1	DBMS	11	RNSIT	2017-07-12

1 row in set (0.0007 sec)

```
update View_for_Student_Lib_DB
set due_date = '2017-05-12'
where card_no = 104;
```

```
select *
from View_for_Student_Lib_DB
where card_no=104;
select * from book_loans
where card_no=104;
```

```
# delete a View
drop view View_for_Student_Lib_DB;
show full tables;
```

```
+-----+-----+
| Tables_in_library_db_2025_26 | Table_type |
+-----+-----+
| book                          | BASE TABLE |
| book_authors                  | BASE TABLE |
| book_copies                   | BASE TABLE |
| book_loans                    | BASE TABLE |
| borrower                      | BASE TABLE |
| library_branch                | BASE TABLE |
| publisher                     | BASE TABLE |
+-----+-----+
7 rows in set (0.0012 sec)
```

Experiment No. 10: Group B - 04

To create a stored Function and its use in MySQL.

```
use Library_DB_2025_26;
select database();
```

Function

The function is used to display number of publishers from a
particular city

```
delimiter $$
create function publisher_count(publisher_address_input
varchar(30))
    returns integer
    deterministic
    begin
        declare p_count integer;
        select count(*) into p_count
        from publisher
        where publisher.address = publisher_address_input;
        return p_count;
    end
$$
delimiter ;
```

```
show function status;
```

```
select publisher_count ('New Delhi') as Publisher_Count;
+-----+
| Publisher_Count |
+-----+
|                3 |
+-----+
1 row in set (0.0008 sec)
```

```
select publisher_count ('Mumbai') as Publisher_Count;
+-----+
| Publisher_Count |
+-----+
|                0 |
+-----+
1 row in set (0.0006 sec)
```

```
select publisher_count ('Bangalore') as Publisher_Count;
```

```
+-----+
```

```
| Publisher_Count |
```

```
+-----+
```

```
|                2 |
```

```
+-----+
```

```
1 row in set (0.0005 sec)
```

```
drop function publisher_count;
```

```
show function status;
```

Experiment No. 11: Group B - 05

To create a stored Procedure and its use in MySQL, and
demonstrate its use using CALL keyword.

```
use Library_DB_2025_26;  
select database();
```

Procedure

To define the procedure that returns the count of publishers

```
delimiter $$
```

```
create procedure publisher_count_procedure(in  
publisher_address_input varchar(30), out p_count int)  
begin  
    select count(*) into p_count  
    from publisher  
    where publisher.address = publisher_address_input;
```

```
end $$  
delimiter ;
```

```
show procedure status;
```

Now call the procedure

```
call publisher_count_procedure ('New Delhi', @p_count);  
select @p_count as Publisher_Count;
```

```
+-----+  
| Publisher_Count |  
+-----+  
|                3 |  
+-----+  
1 row in set (0.0003 sec)
```

```
call publisher_count_procedure ('Mumbai', @p_count);  
select @p_count as Publisher_Count;
```

```
+-----+  
| Publisher_Count |  
+-----+  
|                0 |  
+-----+  
1 row in set (0.0004 sec)
```



```
call publisher_count_procedure ('Bangalore', @pub_count);
select @pub_count as Publisher_Count;
```

```
+-----+
| Publisher_Count |
+-----+
|                2 |
+-----+
1 row in set (0.0004 sec)
```

```
drop procedure publisher_count_procedure;
show procedure status;
```

Expt No 12(A-05) :- Screen design and Report generation: Sample forms and reports should be generate d using any front end tools.

```
import java.sql.* ;
import java.io.*;
class Mysqldb
{
    public static void main(String[] args)
    {
        int ch,id;
        String name,update,update_db;
        PreparedStatement ps;
        try
        {
            Class.forName("com.mysql.jdbc.Driver");
            String url="jdbc:mysql://localhost:3306/rms";
            String uname="root";
            String pass="vicky@123";
            Connection con=DriverManager.getConnection(url,uname,pass);
            InputStreamReader isr=new InputStreamReader(System.in);
            BufferedReader br=new BufferedReader(isr);
            do
            {
                System.out.println("\n 1.Select \t 2.Insert \n 3.Delete \t 4.Update \n 5.Exit");
                System.out.println("Enter your choice");
                ch=Integer.parseInt(br.readLine());

                switch(ch)
                {
                    case 1:
                        String select="select * from ticket1";
                        Statement stmt=con.createStatement();
                        ResultSet rs=stmt.executeQuery(select);
                        while(rs.next())
                        {
                            System.out.println(rs.getInt(1)+"\t"+rs.getString(2)+"\t\t"+rs.getInt(3)+" "+rs.getInt(4));
                        }
                        break;

                    case 2:
                        System.out.println("Enter PNR number");
                        id=Integer.parseInt(br.readLine());
                        System.out.println("Enter Name");
                        name=br.readLine();
                        System.out.println("Enter Train number");
                        int n=Integer.parseInt(br.readLine());
                        System.out.println("Enter mobile number");
                        int mob1=Integer.parseInt(br.readLine());
                        String insert="Insert into ticket1 values(?,?,?,?)";
                        ps=con.prepareStatement(insert);
                        ps.setInt(1,id);
                        ps.setString(2,name);
                        ps.setInt(3,n);
                        ps.setInt(4,mob1);
```

```

ps.executeUpdate();
break;

case 3:
System.out.println("Enter PNR whose record you want to delete");
id=Integer.parseInt(br.readLine());
String delete="delete from ticket1 where PNR=?";
ps=con.prepareStatement(delete);
ps.setInt(1,id);
ps.executeUpdate();
break;

case 4:
System.out.println("\n Update name or mobile no. ?");
update_db=br.readLine();

if(update_db.equals("name"))
{
System.out.println("\n Enter the PNR no. of the person whose name is to be modified ");
id=Integer.parseInt(br.readLine());
System.out.println("\n Enter new name");
name=br.readLine();
update="update ticket1 set name=? where PNR=?";
ps=con.prepareStatement(update);
ps.setString(1,name);
ps.setInt(2,id);
ps.executeUpdate();
}
else if(update_db.equals("mobile no. "))
{
System.out.println("\n Enter the PNR of the person whose mobile no. is to be modified ");
id=Integer.parseInt(br.readLine());
System.out.println("\nEnter new mobile number");
int mob=Integer.parseInt(br.readLine());
System.out.println(mob);
update="update ticket1 set Mob_No=? where PNR=?";
ps=con.prepareStatement(update);
ps.setInt(1,mob);
ps.setInt(2,id);
ps.executeUpdate();
}
break;

case 5: System.exit(0);
}
}while(ch<5);
}
catch(Exception e)
{
System.out.println(e);
}
}
}
}

```

/*Output :-

vinit@vinit-Inspiron-660s:~\$ export CLASSPATH=\$CLASSPATH:mysql-connector-java-5.1.38-bin.jar

vinit@vinit-Inspiron-660s:~\$ javac Mysqldb.java
vinit@vinit-Inspiron-660s:~\$ java Mysqldb

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

1

123456 Vicky 12139 890123

123478 Tony 12465 895632

213445 Nick 21987 421579

456123 Steve 21867 528100

657763 Phil 21987 990232

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

2

Enter PNR number

872392

Enter Name

Barry

Enter Train number

51672

Enter mobile number

754842

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

1

123456 Vicky 12139 890123

123478 Tony 12465 895632

213445 Nick 21987 421579

456123 Steve 21867 528100

657763 Phil 21987 990232

872392 Barry 51672 754842

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

3

Enter PNR whose record you want to delete

657763

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

1

123456 Vicky 12139 890123

123478 Tony 12465 895632
213445 Nick 21987 421579
456123 Steve 21867 528100
872392 Barry 51672 754842

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

4

Update name or mobile no. ?
name

Enter the PNR no. of the person whose name is to be modified
123456

Enter new name
Oliver

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

4

Update name or mobile no. ?
mobile no.

Enter the PNR of the person whose mobile no. is to be modified
123478

Enter new mobile number
602917

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

1

123456 Oliver 12139 890123
123478 Tony 12465 602917
213445 Nick 21987 421579
456123 Steve 21867 528100
872392 Barry 51672 754842

1.Select 2.Insert
3.Delete 4.Update
5.Exit

Enter your choice

5

vinit@vinit-Inspiron-660s:~\$

*/